REGION-BASED LICENSE PLATE LOCALIZATION USING MORPHOLOGICAL AND EDGE-BASED PROCESSING

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Abstract: License plate localization is a crucial preprocessing stage in Automatic Number Plate Recognition (ANPR) systems, as inaccurate localization directly affects overall system performance. Many existing approaches rely on complex machine learning models that require large training datasets, feature extraction, and high computational resources. This project investigates a lightweight, classical image processing–based approach for license plate localization using simple preprocessing filters implemented in Java. The proposed method applies median filtering for noise reduction, grayscale conversion, histogram equalization for contrast enhancement, image binarization, edge detection, and morphological operations to enhance plate visibility. Connected component analysis and geometric constraints, such as aspect ratio filtering, are then used to extract candidate license plate regions. The system is evaluated using a dataset of vehicle images, and localization performance is measured based on detection accuracy. Experimental results demonstrate that enhancing classical preprocessing steps improves license plate localization reliability while maintaining low computational complexity, making the approach suitable for real-world applications without reliance on complex machine learning models.

Keywords**:** Automatic Number Plate Recognition, number plate recognition, localization, image processing, Java programming

1.0 INTRODUCTION

License plate localization is a critical preprocessing stage in Automatic Number Plate Recognition (ANPR) systems, as the accuracy of subsequent recognition stages depends heavily on the correct identification of the license plate region. However, many existing license plate localization systems depend on high-quality camera hardware and hardware-assisted solutions to obtain clear plate images. Such hardware dependency increases system cost, computational complexity, and deployment constraints, making these approaches less suitable for low-cost or lightweight ANPR implementations.

In addition, vehicle images captured in real-world environments frequently contain noise, uneven illumination, and irrelevant background objects, such as vehicle body textures and surrounding structures. When traditional edge detection techniques are applied in isolation, these factors cause the generation of multiple false candidate regions, significantly reducing localization reliability. As a result, a simple and automatic image-processing-based approach is required to accurately localize license plates without relying on specialized hardware or complex learning models, while maintaining low computational complexity.

Previous studies have demonstrated that classical image processing techniques can effectively address these challenges. In particular, Zhai, Benssali, and Ramalingam proposed a license plate localization method based on morphological operations combined with edge information, where the structural characteristics of license plates are enhanced while irrelevant background regions are suppressed. Their approach uses morphological filtering to connect fragmented edges and eliminate noise, followed by geometric analysis to identify valid license plate regions. The results of their work indicate that morphological processing significantly reduces false detections produced by edge-based methods alone, supporting the suitability of region-based techniques for license plate localization in resource-constrained environments.

Motivated by these findings, the present study proposes a region-based license plate localization approach using morphological and edge-based processing implemented in Java. The proposed method applies a sequence of classical preprocessing steps, including noise removal, grayscale conversion, contrast enhancement, edge detection, and morphological filtering, to enhance plate structures while suppressing background interference. Connected component analysis is then performed on the morphologically processed binary image, and geometric constraints such as aspect ratio and region area are applied to accurately identify and extract the true license plate region.

The proposed approach is evaluated using a dataset of vehicle images provided by the lecturer to ensure consistent experimental conditions. The experimental results demonstrate that the region-based method effectively reduces false candidate regions generated by edge detection alone and achieves reliable license plate localization without requiring high-quality camera hardware or computationally expensive models. By relying

solely on classical image processing techniques, the proposed system maintains low computational complexity while providing accurate localization, making it suitable for practical and lightweight ANPR applications.

2.0 METHODS

2.1 IMAGE ACQUISITION

The dataset used for this mini project consists of 97 vehicle images provided by the lecturer to ensure consistency and uniformity in the experimental evaluation.

Figure 1. Dataset

2.2 IMAGE PRE-PROCESSING

In this stage, we aim to enhance the quality of vehicle images that is to prepare the images to highlight important features of the images so the license plate can be detected more reliably. In this project, preprocessing consists of 5 main steps which are grayscale conversion, noise reduction using median filter, edge detection using Sobel operator, image binarization using threshold and morphology.

2.2.1 CONVERSION TO GRAYSCALE

The first preprocessing step is grayscale conversion. The input vehicle image is originally in RGB colour format, which contains three colour channels which are red, green and blue. Processing colour images directly increases computational cost and complexity. For number plate detection in our case, colours are unnecessary for edge and shape analysis. Therefore, the image is converted into a single-channel 8 bit grayscale image.

In the implementation, each pixel's grayscale value is computed using a weighted sum of red, green and blue components. The formula used is:

$$I(x,y) = 0.2126R + 0.7152G + 0.0722B$$

Where R, G and B represent red, green and blue pixel values respectively. More weight is given to the green channel because human eyes are sensitive to green light. The grayscale image represents pixel intensity values ranging from black to where and serves as a simplified input for subsequent processing steps.

2.2.2 NOISE REDUCTION

Noise reduction is applied to remove unwanted disturbances such as salt and pepper effect and small intensity variations. Noise can negatively affect the edge detection by creating false edges. To address this issue, a median filter is used.

After grayscale conversion, the output of the image will be applied median filter. The median filter operates on a 3 x 3 neighbourhood around each pixel. For each pixel position, the intensity values within the neighbourhood are collected, sorted in ascending order and he middle value which is the median value is selected as the new pixel value. This method is effective in reducing noise while preserving the important edges of the image which are essential for accurate license plate boundary detection.

2.2.3 EDGE DETECTION

Once the image has been smoothed using median filter, edge detection is performed to highlight the areas where there are strong intensity changes. These edges usually correspond to object boundaries, including the borders of the license plate.

The Sobel operator is used for edge detection in this system. It consists of two 3 x 3 convolutional masks:

$$Gx = [-1\ 0\ 1;\ -2\ 0\ 2;\ -1\ 0\ 1]$$

$$Gy = [1\ 2\ 1;\ 0\ 0\ 0;\ -1\ -2\ -1]$$

where Gx for detecting horizontal edges and Gy is for detecting vertical edges The image is convolved with both masks to calculate the horizontal and vertical gradients. The resulting edge image emphasizes the structural features on the vehicle and the license plate regions.

2.2.4 BINARIZATION

After edge detection, the image will be converted into a binary image to separate significant edge information from the background. Instead of using fixed threshold value, an adaptive thresholding approach based on histogram analysis is applied to improve the plate localization

In this project, Otsu's thresholding method is used to automatically determine an optimal threshold value from the image histogram. The method analyses the distribution of pixel intensities and selects the threshold that maximizes the separation between foreground and background regions. This allows the binarization process to adapt to the differences in images across the dataset.

The binarized image highlights strong edge structure while suppressing weak or irrelevant edges. This step is essential because subsequent morphological processing operate on binary images. By using an adaptive thresholding rather than a fixed one, the binarization stage improves the reliability of license plate localization.

2.2.5 MORPHOLOGICAL

Morphological processing is applied to the binarized edge image to enforce spatial connectivity and improve the structural integrity of candidate license plate regions (Vijayalakshmi et al., 2020). After edge detection and global thresholding, the binary image contains fragmented edge components due to noise and illumination variation which prevents the license plate from forming a single connected region suitable for localization.

In this project, a binary morphological closing operation is employed, defined as $I \cdot B = (I \oplus B) \ominus B$ where I denotes the binarized image and B is a 3 x 3 structuring element. The dilation operation $(I \oplus B)$ assigns each pixel the maximum value within its local neighbourhood, as implemented by selecting the maximum intensity in a 3 x 3 window (Hermary et al., 2022). This expands foreground regions and bridges small gaps between adjacent edge segments and character strokes. The erosion operation ($I \ominus B$) subsequently assigns the minimum neighbourhood value, removing spurious expansions introduced during dilation while preserving overall region connectivity.

Morphological closing is applied after binarization because morphological operators are defined for binary images and rely on explicit foreground-background separation. This operation consolidates fragmented

plate features into a single connected component, enabling reliable region-based segmentation and subsequent geometric filtering based on area and aspect ratio.

2.3 NUMBER PLATE LOCALIZATION

After morphological processing, the license plate is localized using a region-based segmentation and geometric filtering approach. At this stage, the morphologically processed binary image contains multiple connected foreground regions corresponding to various objects in the scene, such as vehicle edges, background structures and the license plate. The objective of the localization stage is to identify which of these regions corresponds to the license plate. Region-based segmentation is performed by scanning the binary image and grouping connected foreground pixels into distinct regions using an 8-connected neighbourhood. Each connected region is treated as a candidate object and a tight bounding box is computed by tracking the minimum and maximum pixel coordinates in both horizontal and vertical directions. This process transforms pixel-level information into region-level representations that can be quantitatively analysed.

To distinguish the license plate from other connected regions, geometric feature filtering is applied. For each candidate region, basic shape descriptors are extracted from its bounding box, specifically the area and aspect ratio (defined as the ratio of width to height). These features are selected because license plates exhibit a characteristic rectangular shape with a relatively large width compared to height. Regions that do not satisfy predefined constraints on area and aspect ratio are discarded while the remaining region that best matches these geometric characteristics is selected as the license plate (Jadhav et al., 2022). Once the license plate region is identified, its bounding box is projected onto the original colour image for visualization. This step does not influence the detection process itself but provides a clear and interpretable output, highlighting the localized license plate region in the original image.

FINDINGS AND ARGUMENT

Figure 1. Combined preprocessing output showing the result after grayscale conversion, noise reduction, edge detection, binarization and morphological closing.

The experimental results demonstrate that the proposed classical image processing–based approach is capable of localizing license plates under varied imaging conditions. Figure 1 illustrates the intermediate output obtained after applying the combined preprocessing stages, including grayscale conversion, noise

reduction, edge detection, binarization and morphological closing. This image shows that fragmented edge structures are successfully consolidated into coherent regions which are suitable for subsequent region-based segmentation and geometric analysis.

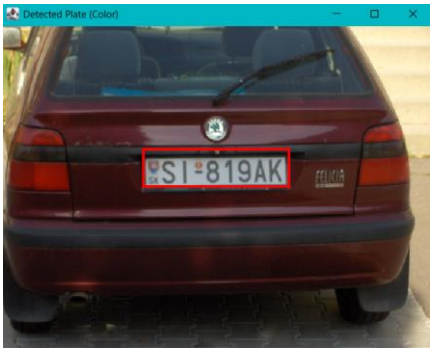

Figure 2. Final output showing the localized license plate highlighted with a bounding box on the original color image.

The final detection result is shown in Figure 2 where the selected license plate region is highlighted using a bounding box overlaid on the original color image. This result confirms that the region-based segmentation and geometric filtering stages are effective in isolating the license plate from other background objects. The bounding box accurately encloses the plate area, indicating that the chosen geometric constraints, namely area and aspect ratio are appropriate for license plate localization.

To quantitatively evaluate the system, detection accuracy was measured using the provided dataset. Out of 97 test images, the proposed method successfully localized the license plate in 72% of the images. This result indicates that the system performs reasonably well despite relying solely on classical image processing techniques without the use of machine learning or deep learning models. Detection failures were primarily observed in images with severe illumination variation, low contrast or highly cluttered backgrounds.

In terms of computational efficiency, the processing time for a single test image was measured in milliseconds. For the test case shown in Figure 2, the complete license plate localization pipeline required 967.15 ms (about 2 seconds) to process one image. Although the system relies solely on classical image processing techniques without hardware acceleration or learning-based optimization, the execution time demonstrates that the proposed method is computationally efficient and suitable for near real-time applications particularly in environments with limited computational resources.

CONCLUSIONS

This study demonstrates that region-based license plate localization using classical morphological and edge-based image processing techniques is an effective and computationally efficient approach for Automatic Number Plate Recognition (ANPR) systems. Experimental results show that a structured preprocessing pipeline comprising noise reduction, grayscale conversion, contrast enhancement, edge detection, binarization, and morphological closing successfully consolidates fragmented plate features into coherent regions. The combination of connected component analysis with geometric constraints, such as aspect ratio and region area, enables reliable discrimination of license plate regions from background objects without relying on complex machine learning models.

Despite its effectiveness, several limitations were observed. The proposed method shows reduced performance under severe illumination variation, low-contrast conditions, or highly cluttered backgrounds, where edge information becomes weak or inconsistent. Furthermore, the use of fixed thresholds and predefined geometric constraints limits adaptability to diverse plate formats and imaging conditions, highlighting the sensitivity of purely classical approaches to environmental variability.

From a practical perspective, the proposed Java-based implementation provides a lightweight and easily deployable solution suitable for systems with limited computational resources, achieving near real-time performance using only classical image processing techniques. Theoretically, this work reinforces the

relevance of region-based segmentation and morphological processing as effective tools for object localization tasks involving well-defined geometric structures.

In conclusion, the proposed approach offers reliable license plate localization with low computational complexity, making it suitable as a baseline for lightweight ANPR applications. Although it does not achieve the robustness of learning-based methods in all scenarios, it provides a solid foundation for future improvements, such as adaptive thresholding, enhanced contrast processing, or the integration of lightweight classification methods to improve robustness under challenging conditions.

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